

Chapter 53: Chronic Hypertension — Revision Notes

Williams Obstetrics 26e, pp. 2457–2488 of the PDF. High-yield numbers are in **bold**. The six tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 3 figures are included from the PDF.

Big picture: Hypertension affects **~6%** of reproductive-age women and complicates **1.8–2.3%** of pregnancies. BP usually falls early in pregnancy, then rises in the third trimester. The dominant danger is **superimposed preeclampsia (20–50%)**, which drives abruption, FGR, stillbirth and maternal death. **Severe hypertension ($\geq 160/110$) must be treated**; the benefit of treating mild–moderate disease is unproven. ACE inhibitors and ARBs are contraindicated.

1. General Considerations

- BP is a **polygenic** trait (plus epigenetics); rises with age and weight. Normal range in adults is broad → defining hypertension is difficult.

Definition and classification

- 140/90 mm Hg** is the usual upper limit of normal (derived from actuarial data on white men).
- Incidence of BP >140/90: **blacks 34%, whites 29%, Mexican Americans 21%**.
- New ACC/AHA criteria (Whelton, 2018) differ from the pregnancy (ACOG) criteria:

Table 53-1. Criteria for Diagnosis of Hypertension

SBP (mm Hg)		DBP (mm Hg)	Nonpregnant (ACC/AHA)	Pregnant (ACOG)
<120	and	<80	Normal	Normal
120–129	and	<80	Elevated	Normal
130–139	or	80–89	Stage 1 HTN	Normal
140–159	or	≥ 90	Stage 2 HTN	Mild to moderate HTN
≥ 160	or	≥ 110	Stage 2 HTN	Severe HTN

ACC = American College of Cardiology; ACOG = American College of Obstetricians and Gynecologists; AHA = American Heart Association; DBP = diastolic blood pressure; HTN = hypertension; SBP = systolic blood pressure.

Memory hook: In pregnancy there are only three bins — **<140/90 normal, 140–159/90–109 mild–moderate, $\geq 160/110$ severe**. The ACC/AHA "stage 1" (130–139/80–89) counts as normal for ACOG.

Treatment and benefits in nonpregnant adults

- Hypertension causes **~15% of deaths worldwide**; ↑ CVD, CHD, heart failure, stroke, renal failure, PAD and death — all ↓ with treatment.
- Nonpregnant guidelines treat **stage 1** if CVD risk factors. The best approach before and during pregnancy is **unknown**.

Preconceptional counseling

- Ask about duration, control and current drugs; **check home BP devices** for accuracy; assess health, activity and diet (Table 53-2).
- Multiple drugs or poor control → higher risk.**

- **Hypertension >5 years or with diabetes** → **assess cardiac and renal function**: serum creatinine + **spot urine protein:creatinine ratio**; if **>0.3** → quantify with 24-h urine.
- Prior stroke, MI, arrhythmia or ventricular failure → markedly ↑ risk of recurrence/worsening.
- No consensus that poor control contraindicates pregnancy, but **counsel about marked risks** if:
 - persistent SBP **≥160** or DBP **≥110** despite therapy
 - **multiple antihypertensives** needed
 - serum **creatinine >2 mg/dL**
 - prior **stroke, MI or cardiac failure**

Table 53-2. Lifestyle Modifications for Hypertensive Patients

Modification	Recommendation
Weight reduction	BMI ≤25 kg/m²
Diet	Emphasize vegetables, fruits, whole grains; include low-fat dairy, poultry, fish, legumes, nontropical vegetable oils, nuts; limit sweets and red meats. Examples: DASH , USDA Food Pattern, AHA Diet
Lower sodium intake	No more than 2400 mg/d; 1500 mg/d desirable
Aerobic physical activity	3–4 sessions/wk, ~40 min each , moderate- to vigorous-intensity
Alcohol moderation	≤1 drink daily (none when pregnant)

AHA = American Heart Association; DASH = Dietary Approaches to Stop Hypertension; USDA = United States Department of Agriculture. Summarized from Kotchen, 2018; Whelton, 2018.

2. Pregnancy Considerations

Diagnosis and evaluation

- **Chronic hypertension** = hypertension preceding pregnancy or identified before **20 wk** (preeclampsia is rare before 20 wk).
- ACOG threshold: **SBP ≥140 or DBP ≥90 mm Hg**.
- Suspected **white-coat hypertension** → serial ambulatory monitoring.
- Repeated **gestational hypertension or prior preeclampsia** (especially **early-onset**) → risk factor for **latent chronic hypertension**. Gestational hypertension is analogous to gestational diabetes (a chronic hypertensive diathesis).
- **Secondary causes** to consider: chronic renal disease, **obstructive sleep apnea**, connective tissue disease, **primary aldosteronism**, **Cushing syndrome**, **pheochromocytoma**. Most cases are uncomplicated essential hypertension.

Risk factors

- Most cited: **ethnicity (black women)**, **obesity**, **diabetes**.
- **Obesity may raise chronic hypertension prevalence 10×** and ↑ superimposed preeclampsia.
- **Metabolic syndrome** = risk marker for superimposed preeclampsia and persistent postpartum hypertension.
- Commonest comorbidities: **pregestational diabetes 6.6%**, thyroid disorders 4.1%, collagen vascular disease 0.6%.

Physiological effects

- **1st trimester: BP falls** because of a **30% ↓ in SVR**; then BP **rises in the 3rd trimester** in most chronically hypertensive women.

- Persistently ↑ vascular resistance and possibly ↓ plasma volume expansion; arterial stiffness most marked with superimposed preeclampsia.

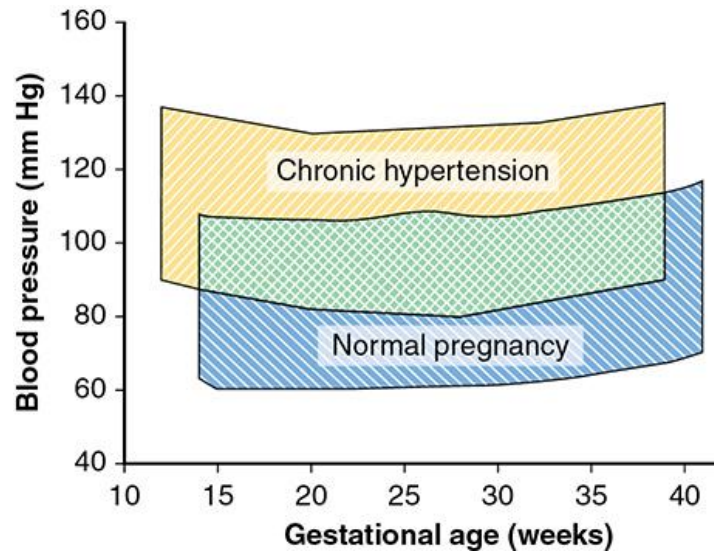


Figure 53-1. Chronic hypertension vs normal pregnancy: both show a midpregnancy dip and a third-trimester rise, with the chronic hypertensive curve running higher throughout.

3. Adverse Pregnancy Effects

- Complication rates rise with **severity and duration** of prepregnancy hypertension. ACC/AHA "elevated" BP may also predict adverse outcomes.
- **Superimposed preeclampsia, especially early-onset, is especially dangerous.**

Table 53-3. Some Adverse Effects of Chronic Hypertension on Maternal and Perinatal Outcomes

Maternal	Perinatal
Superimposed preeclampsia	Stillbirth
HELLP syndrome	Fetal-growth restriction
Placental abruption	Preterm delivery
Pulmonary edema	Neonatal death
Stroke	Neonatal morbidity
Acute kidney injury	Congenital anomalies
Heart failure	
Hypertensive cardiomyopathy	
Myocardial infarction	
Maternal death	

HELLP = hemolysis, elevated liver enzyme levels, low platelet count.

Maternal morbidity and mortality

- **Well-controlled hypertension without end-organ damage → usually does well.**
- Complications more likely with severe baseline hypertension and end-organ damage — especially **LV hypertrophy** and **hypertensive glomerulosclerosis**.
- Nationwide sample (per 1000): **pulmonary edema 1.5, stroke 2.7, mechanical ventilation 3.8, AKI 5.9, maternal death 0.4.**

- Also: peripartum cardiomyopathy, coronary and aortic dissection.
- ⚠️ **SBP ≥ 160 or DBP ≥ 110 rapidly causes renal or cardiopulmonary dysfunction or cerebral hemorrhage.**
- Superimposed preeclampsia/eclampsia $\rightarrow$ poor maternal prognosis unless pregnancy is ended.
- **Chronic hypertension $\rightarrow$ 5 $\times$ maternal death risk.** Hypertensive disorders = **7.4%** of US pregnancy-related deaths (2011–2013); related causes: CV conditions 15.5%, cardiomyopathy 11%, stroke 6.6%.

Superimposed preeclampsia

- **Incidence 20–50%** (imprecisely defined); **25%** in the MFMU trial.
- Risk $\uparrow$ with **severity of baseline hypertension** and **need for antihypertensives**; greater still with **baseline proteinuria** (end-organ damage).
- **BP $\geq 130/80$ early in pregnancy $\rightarrow$ excess preeclampsia.**
- Women who develop preeclampsia have higher MAP throughout and an **earlier MAP nadir (23.3 vs 26.4 wk).**

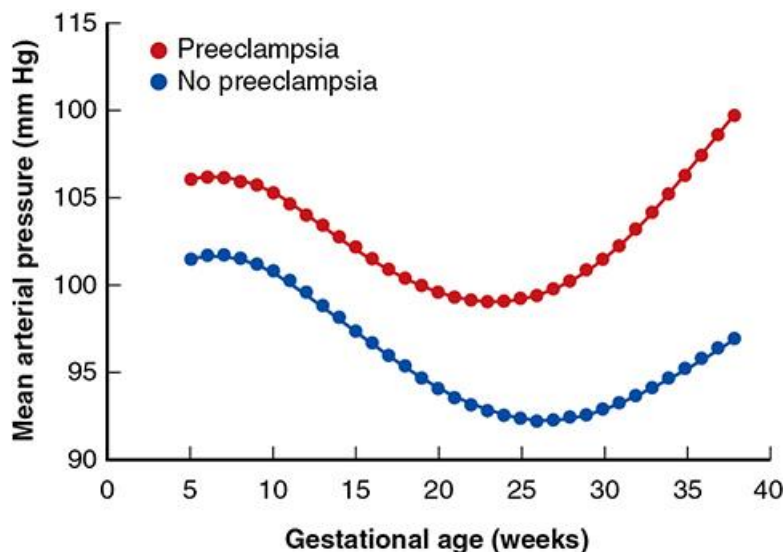


Figure 53-2. Treated chronic hypertension: women who develop superimposed preeclampsia have higher MAP from the start and an earlier nadir (~23 vs ~26 wk).

Prediction — individual tests disappointing:

- First-trimester **inhibin A** $\downarrow$ (low sensitivity, not useful) and **PAPP-A** $\downarrow$.
- **Low PIGF, high sFlt-1** associated with superimposed preeclampsia.
- $\uparrow$ First-trimester **umbilical artery pulsatility index** (as stated in the text).
- **Combining** factors may help.

Prevention

- Most drugs show little or no benefit. **Low-dose aspirin** is best studied: USPSTF "moderate certainty" of benefit.
- **ACOG: aspirin 81 mg/d, started between 12 and 28 wk, continued until delivery.**
- **Vitamins C and E do not work** (17% vs 20% preeclampsia).

Placental abruption

- **2–3 $\times$ $\uparrow$ risk:** general population **1 in 200–300** $\rightarrow$ chronic hypertension **1 in 60–120**.
- Mostly with worsening hypertension or superimposed preeclampsia; **8.4%** with severe hypertension.

Perinatal morbidity and mortality

- Almost all adverse perinatal outcomes $\uparrow$; much higher with preeclampsia; rates rise **stepwise with BP** (Fig. 53-3).
- **Congenital anomalies** (treated or untreated): **cardiac defects OR 1.5** (RR 1.4); **esophageal atresia aOR 3.2** in untreated hypertension.
- **Stillbirth 15–24 per 1000** (superimposed preeclampsia, abruption, FGR).

- **FGR averages 20%** (2.5× in Swedish study; 1.3× in Dutch nulliparas). With superimposed preeclampsia **~50% vs 21%** without. Treated hypertension → **11%** with birthweight **≤3rd percentile**.
- **Perinatal mortality 2–4× ↑**. By severity: **mild 31, moderate 72, severe 100 per 1000**. Parkland (treated): **32 per 1000**. **Doubles with superimposed preeclampsia**. Worse still with coexisting diabetes.
- Offspring: long-term endocrine/metabolic morbidity, especially **obesity**.

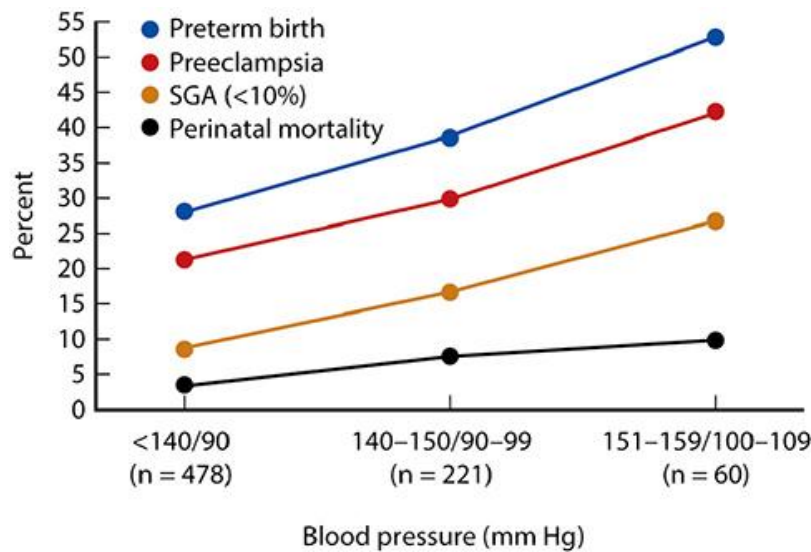


Figure 53-3. Mild chronic hypertension: preterm birth, preeclampsia, SGA and perinatal mortality all rise with BP category (<140/90 → 151–159/100–109).

4. Management During Pregnancy

- Goals: **prevent moderate/severe hypertension**, delay or dampen superimposed preeclampsia, ↓ adverse outcomes.
- **Home BP self-monitoring** (calibrated devices). Diet counseling; stop tobacco, alcohol, cocaine and other substances. **Low-sodium diet and low-dose aspirin**.
- Long-standing or untreated hypertension → **concentric LVH and proteinuria** concerns → **baseline echocardiogram and 24-h urine protein** early in prenatal care.

Antihypertensive drugs

Table 53-4. Some Antihypertensive Drugs Used for Treatment of Chronic Hypertension During Pregnancy^a

Drug	Comment
Adrenergic-receptor agents^b	
β-blockers: propranolol, metoprolol	Avoid with asthma, decompensated heart function
α/β-blocker: labetalol	—
α-agonist: methyldopa	Less effective for severe hypertension; sedative side effects
Calcium-channel blocking agents^b	
Nifedipine, amlodipine, verapamil	Avoid with tachycardia
Diuretic^c	
Hydrochlorothiazide	Begin before 20 weeks' gestation
Vasodilating agent^c	
Hydralazine	Hypertension complicated by cardiac or renal dysfunction ; less effective with severe hypertension

^aDrugs compatible with pregnancy (Briggs, 2022). ^bFirst-line agents. ^cSecond-line agents.

Adrenergic-receptor blockers

- **β-blockers** (propranolol, metoprolol) ↓ sympathetic tone and CO. ⚠ **Atenolol not recommended — linked with FGR.**
- **Labetalol** (α/β-blocker) is considered safe.
- **Methyldopa and clonidine** act centrally (↓ sympathetic outflow). **Methyldopa and labetalol are the most used in pregnancy.**

Calcium-channel blockers

- **Nifedipine** (dihydropyridine), **verapamil** (phenylalkylamine). Negative inotropes → worsen ventricular dysfunction/heart failure; may theoretically potentiate **magnesium sulfate**.
- **Nifedipine = labetalol in efficacy**, with **less BP variability**. Appear safe.

Diuretics

- Thiazides (sulfonamides) and loop diuretics: short-term Na⁺/water diuresis → later "sodium escape"; long-term benefit partly from ↓ PVR.
- Mildly **diabetogenic**; may curtail normal **plasma volume expansion** → not first-line, especially **after 20 wk**. Overall **thiazides are considered safe**.

Vasodilators

- **Hydralazine**: IV for severe peripartum hypertension for decades. **Oral monotherapy not used** (weak, reflex tachycardia); useful **adjunct** with renal insufficiency or ventricular dysfunction.

ACE inhibitors and ARBs ⚠

- ACE inhibitors in the **2nd and 3rd trimesters** → **oligohydramnios, hypocalvaria, renal dysfunction**; possible 1st-trimester teratogenicity.
- **Not recommended at any time in pregnancy. ARBs presumed to have the same effects → contraindicated.**

Antihypertensive treatment in pregnancy

Mild or moderate hypertension

- Whether to start or continue therapy is **debatable**.

- Cochrane (~5000 women): treatment ↓ **progression to severe hypertension**, but **no difference** in superimposed preeclampsia, preterm birth, SGA or perinatal mortality.
- Lowering MAP may ↓ uteroplacental perfusion → **SGA** (von Dadelszen; conflicting data). Large RCTs: no ↑ FGR with treatment. Worsening BP itself impairs growth. **Unresolved.**

Severe chronic hypertension

- **SBP ≥160 or DBP ≥110** → **treat** (↓ maternal morbidity and mortality).
- With **cardiovascular or renal end-organ damage**, treat at **SBP ≥140–150 or DBP ≥90–100**.
- BP **≥170/110 at 6–11 wk** (44 pregnancies; methyldopa + hydralazine to keep <160/110): **½ developed superimposed preeclampsia**, and **all adverse perinatal outcomes occurred in that half**. Confirmed in 447 treated women (Parkland).
- **Baseline proteinuria >300 mg/d** → more superimposed preeclampsia, preterm birth and SGA (Table 53-5). **LVH/abnormal remodeling** → preeclampsia, preterm birth, NICU admission.

Table 53-5. Selected Pregnancy Outcomes in Women with Chronic Hypertension with and without Baseline Proteinuria^a and Who Were Treated During Pregnancy

Outcome	Baseline Proteinuria ^a	No Proteinuria	p value
Superimposed preeclampsia	79%	49%	<0.001
Abruption	0	1%	0.45
EGA at delivery (mean) ^b	35.1 ± 4.3 wk	37.2 ± 3.3 wk	<0.001
≤30 weeks	18%	6%	0.001
≤34 weeks	34%	17%	0.005
≤37 weeks	48%	26%	0.002
Birthweight (mean) ^b	2379 ± 1028 g	2814 ± 807 g	<0.001
≤3rd percentile	20%	9%	0.01
≤10th percentile	41%	22%	<0.001
Perinatal mortality	36/1000	31/1000	0.47

^aDefined as ≥300 mg/d protein excretion before 20 weeks' gestation. ^bMean ± standard deviations. EGA = estimated gestational age.

"Tight control"

- Observational data (Ankumah) suggested better outcomes with BP <140 before 20 wk — **not confirmed**.
- **CHIPS trial (Magee, 987 women)**: tight vs less-tight control → **only severe hypertension ↓ (28% vs 41%)**; nothing else differed. No gestational age at which tight control was better.
- **ACOG: no proven benefit or harm** of "tight" (target DBP <85) vs "less-tight" (target DBP <100). CHAP trial ongoing (at the time of writing).

Table 53-6. Selected Maternal and Perinatal Outcomes in Pregnant Women with Chronic Hypertension According to Less-Tight versus Tight Control

Outcome	Less-Tight Control	Tight Control
Maternal		
Placental abruption	2.2%	2.3%
Severe hypertension^a	41%	28%
Preeclampsia	49%	46%
HELLP syndrome	1.8%	0.4%
Perinatal		
Deaths	28/1000	23/1000
Weight <10th percentile	16%	20%
Weight <3rd percentile	4.7%	5.3%
Respiratory problem	17%	14%

^a $p < 0.001$; all other comparisons $p > 0.05$. HELLP = hemolysis, elevated liver enzyme levels, low platelet count.

Recommendations for therapy (ACOG / SMFM)

- **Must treat:** severe hypertension (neuro-, cardio- and renoprotection); prior **stroke, MI, or cardiac/renal dysfunction**.
- Reasonable to start in healthy women with persistent **SBP >150 or DBP >95–100**.
- **Parkland: start at $\geq 150/100$ mm Hg.** Preferred monotherapy: **labetalol**, or a calcium-channel blocker (**nifedipine or amlodipine**).
- **Thiazide adjunct** reasonable in the **first half** of pregnancy — especially in **black women** (salt-sensitive hypertension).
- **Already on drugs at booking:**
 - ACOG/SMFM: for mild–moderate hypertension, reasonable to **stop in the 1st trimester** and restart if BP approaches the severe range.
 - Parkland: **continue** therapy.
 - **Always stop ACE inhibitors and ARBs.**
- **Refractory hypertension** → think of **superimposed preeclampsia**, inaccurate measurement, suboptimal treatment, **illicit drugs**, chronic **NSAID** use.

Superimposed preeclampsia — diagnosis

- Supporting features: **worsening hypertension, new-onset proteinuria**, severe headache, visual disturbance, generalized edema, oliguria, convulsions, pulmonary edema.
- **Pitfalls:**
 1. BP normally rises near the **end of the 2nd trimester** in chronic hypertension (the upper end of Fig. 53-1) → if preeclampsia is excluded, start or increase treatment.
 2. Pre-existing end-organ damage mimics preeclampsia; **worsening proteinuria is unreliable with underlying renal disease**.
- **Supportive labs:** rising **creatinine** or **transaminases**, **thrombocytopenia**, any feature of **HELLP**.
- **With severe features** → **magnesium sulfate** neuroprophylaxis.

Expectant management of early-onset superimposed preeclampsia

- **40–50%** develop before 37 wk.
- **Maternal outcomes are generally better with delivery**, even if very preterm; delay → abruption, cerebral hemorrhage, peripartum heart failure, and later maternal heart disease.

- 42 of 68 selected women (median 31.6 wk) managed expectantly: **17% abruption or pulmonary edema**, latency **+10 days**, no neonatal benefit.
- ACOG: **<34 wk** may be candidates for expectant management **only where maternal and neonatal ICU resources exist**. Parkland delivers once diagnosed.

Fetal assessment

- **Antepartum surveillance recommended** (↑ stillbirth, FGR); ideal test, interval and start are unknown.
- **All: third-trimester sonographic growth assessment.**
- **Low-risk** (Chahine) = all four of:
 1. **Age ≤35 y**
 2. Mild–moderate BP **not requiring medication**
 3. **No prior** hypertension-related adverse pregnancy outcome
 4. **No end-organ damage**
- Low-risk → **weekly NST or BPP**. **High-risk** (fails any criterion) → **earlier sonography and testing**.

Delivery timing (ACOG)

Situation	Timing
Mild–moderate, uncomplicated, on medication	Wait until 37–38 wk; not before 37⁰/₇ wk
Mild–moderate, uncomplicated, no medication	Not before 38⁰/₇ wk
Consensus committee	Consider 38–39 wk (≥37 completed weeks)
FGR	Individualize (maternal health vs EFW, GA, neonatal survival)

- Beyond **39 wk** → ↑ severe preeclampsia; planned delivery **<37 wk** → ↑ adverse neonatal outcomes.

Intrapartum considerations

- **Route by obstetric factors**; **induction** preferred and many deliver vaginally. FGR fetuses often need cesarean for nonreassuring FHR.
- **Magnesium sulfate only with superimposed severe preeclampsia.**
- **Epidural ideal** but **not a treatment for hypertension**; women with superimposed preeclampsia are **more sensitive to epidural hypotension**. Superimposed preeclampsia may lengthen the first stage.
- **Severe hypertension (DBP ≥110 or SBP ≥160) → IV hydralazine, IV labetalol or oral nifedipine** (some treat at DBP 100–105).
 - **Hydralazine**: more maternal **palpitations and tachycardia**.
 - **Labetalol**: more **neonatal hypotension and bradycardia**.

Postpartum care

- Manage like severe preeclampsia–eclampsia. **Persistent severe hypertension → consider pheochromocytoma or Cushing disease.**
- With end-organ damage: ↑ **cerebral/pulmonary edema, heart failure, renal dysfunction, cerebral hemorrhage**, especially in the **first 48 h**, often preceded by sudden ↑ MAP (especially systolic).
- Mechanism: ↑ peripheral resistance + mobilized interstitial fluid → ↑ LV workload → diastolic/systolic dysfunction → **pulmonary edema → prompt BP control + furosemide diuresis**.
- Continue the antepartum regimen; untreated women can use **labetalol or nifedipine**.
- **Furosemide** augments postpartum diuresis: **20 mg PO daily × 5 days** helped BP control after severe preeclampsia; daily furosemide ↓ need for antihypertensives at discharge.
- **Daily weights**: a woman should weigh **~15 lb less immediately after delivery**; (last prenatal weight – 15 lb) vs current weight estimates excess extracellular fluid.
- Chronic **NSAID** use may ↑ BP after severe preeclampsia (intermittent PRN use is probably fine).

- Postpartum readmission for severe hypertension **~1.5%**.
- Contraception: discuss before delivery; some methods are less ideal/contraindicated (Ch 38). SMFM encourages **immediate postpartum LARC** in the highest-risk women.

Long-term prognosis

- **High lifetime cardiovascular risk**, especially with diabetes, obesity and metabolic syndrome.
- Persistent hypertension 3 years after severe preeclampsia → **thicker LV septum**.

Quick-Recall: Antihypertensive Drugs in Pregnancy

Drug	Status	Key point
Labetalol	First-line	α/β -blocker, safe; IV for acute severe HTN (neonatal hypotension/bradycardia)
Nifedipine / amlodipine	First-line	As effective as labetalol, less BP variability; avoid with tachycardia; oral nifedipine for acute severe HTN
Methyldopa	First-line	Central α -agonist; sedation; weaker for severe HTN
Propranolol, metoprolol	First-line	Avoid with asthma, decompensated heart
Atenolol	Avoid	FGR
Hydrochlorothiazide	Second-line	Begin before 20 wk; helpful in black women; mildly diabetogenic, ↓ volume expansion
Hydralazine	Second-line (oral)	Oral adjunct for renal/cardiac dysfunction; IV for acute severe HTN (maternal tachycardia)
ACE inhibitors / ARBs	Contraindicated	Oligohydramnios, hypocalvaria, renal dysfunction (2nd–3rd trimester)

Key Numbers to Memorize

- Prevalence **~6%** of reproductive-age women; **1.8–2.3%** of pregnancies
- Chronic HTN = before pregnancy or **<20 wk**; **≥140/90** (mild–moderate), **≥160/110** (severe)
- Counsel on marked risk: BP **≥160/110** on therapy, multiple drugs, creatinine **>2 mg/dL**, prior stroke/MI/heart failure; UPCR **>0.3** → 24-h urine
- SVR ↓ **30%** in 1st trimester
- Maternal death **5×**; hypertensive disorders **7.4%** of pregnancy-related deaths
- Superimposed preeclampsia **20–50%** (MF MU **25%**); **79% with baseline proteinuria vs 49%**; **40–50% occur <37 wk**
- Aspirin **81 mg** from **12–28 wk** until delivery
- Abrupton **1 in 60–120** (vs 1 in 200–300); **8.4%** with severe HTN
- Stillbirth **15–24/1000**; FGR **~20%** (~50% with preeclampsia); perinatal mortality **31 / 72 / 100 per 1000** (mild / moderate / severe)
- Cardiac defects **OR 1.5**; esophageal atresia **aOR 3.2**
- Parkland treats at **≥150/100**; end-organ damage → treat at **≥140–150/90–100**
- Tight (DBP **<85**) vs less-tight (DBP **<100**): only severe HTN ↓ (**28% vs 41%**)
- Delivery: **37⁰/₇** on meds, **38⁰/₇** without; expectant management of superimposed preeclampsia only **<34 wk** with ICU resources
- Postpartum: furosemide **20 mg × 5 days**; weight **–15 lb** after delivery; readmission **~1.5%**; risk window **first 48 h**